Section 5.3: Trigonometric Functions of Any Angle

MATH 130 · Monday, November 9 · Quiz 10 on Wednesday, November 11

Name _____

Reference angle θ′ (to the nearest x-axis): QI θ · QII 180° − θ · QIII θ − 180° · QIV 360° − θ. Outside one rotation, find a coterminal angle first.
Signs (ASTC): QI all positive · QII sin, csc · QIII tan, cot · QIV cos, sec. **From a point (x, y):** $r = \sqrt{(x^2 + y^2)}$, $\sin\theta = y/r$, $\cos\theta = x/r$, $\tan\theta = y/x$.
Exact values: $\sin 30° = \frac{1}{2}$, $\cos 30° = \sqrt{3}/2$ · $\sin 45° = \cos 45° = \sqrt{2}/2$ · $\sin 60° = \sqrt{3}/2$, $\cos 60° = \frac{1}{2}$.

1 · Reference angle, then an exact value

Model (a) θ = 135°: find θ′ and sin θ.
(b) θ = −5π/6: find θ′ and cos θ.

1–2
1. θ = 330°: find θ′ and cos θ.
2. θ = −2π/3: find θ′ and sin θ.

2 · The quadrant from sign conditions

Model sin θ < 0 and tan θ > 0. Which quadrant?

3 tan θ < 0 and sec θ > 0. Which quadrant?

3 · Solve on [0°, 360°): two answers

Find the reference angle from the absolute value, decide which two quadrants have the right sign, then place θ′ in each.

Model sin θ = −½

4–5 **4.** cos θ = √2/2 **5.** sin θ = −√3/2

4 · All six functions from a point

Model The terminal side passes through (5, −12). Find r and all six functions.

6 The terminal side passes through (−6, 8). Find r and all six functions.

Check yourself: a reference angle is always between 0° and 90° and is measured to the x-axis, never the y-axis. *r* is always positive. Each equation in Part 3 has exactly two answers on one rotation.